

Smart Door Lock Face Recognition and RFID Technology for ITDS Department-SC

Apura M. Arben, James Harvey D.V. Baloto,
Zayne David Emmanuel M. Gener, Haydrian C. Tumbagahon,
John Ezrhon C. Victorio, and Elenita T. Capariño, DIT
Bachelor of Science in Information Technology, Bulacan State University,
Sarmiento Campus
San Jose Del Monte City, Bulacan, Philippines
arbenapura.official@gmail.com, jamesharveydevera@gmail.com,
zaynegener@gmail.com, haydriantumbagahon1205@gmail.com,
johnezhonvictorio1022@gmail.com

ABSTRACT

With how quickly technology is developing, it is crucial to secure academic areas. The need for improved security inside Bulacan State University, Sarmiento Campus ITDS faculty is addressed in this study. Traditional locks pose possible risks, such as hardware deterioration, lost keys, and human error. In order to reduce the dangers of unwanted entry and only permit authorized access, we suggest a smart door lock system that combines facial recognition and RFID technology.

This study examines the system's technological needs, implementation difficulties, and advantages, providing guidance for enhancing campus security and creating a secure learning environment that encourages creativity and teamwork.

This study uses the Rapid Application Development (RAD) methodology, an iterative process that ensures efficient prototyping and testing of the system. The system is evaluated using the ISO/IEC 25010 standard, gathering feedback from 28 respondents, including 4 ITDS Professors, 4 IT experts and 20 IT students. The results showed that the system scored an overall mean of 4.59 from ITDS Professors, 4.61 from IT experts, and 4.71 from students, which is equivalent to an "excellent" rating.

1. INTRODUCTION

In an era of fast technological advancement, maintaining security within Academic institutions is crucial (Whitman & Mattord, 2021). In these settings, maintaining the confidentiality of faculty offices is very important since it not only secures valuable assets but also fosters an atmosphere that is favorable to cooperation, instruction, and research projects (Garfinkel, 2017). This requirement is also met by the Department of Information Technology and Data Sciences (ITDS) at Bulacan State University Sarmiento Campus.

Although the ITDS faculty has not yet experienced unauthorized entry, reports from other faculties within the university indicate an increasing likelihood of such incidents, highlighting the fundamental flaws in standard door lock security solutions (Sai et al., 2024). The potential vulnerability of the ITDS faculty to similar incidents underscores the importance of addressing possible security gaps. Manual locking methods introduce the risk of human error and oversight, and instances of lost, stolen, or broken keys could disrupt access control protocols. Additionally, the gradual decay and weakening of locks over time may compromise their effectiveness in

providing security for the faculty room.

Recognizing the importance of preventing such vulnerabilities, there is an opportunity to implement a strong security solution that utilizes state-of-the-art technology. Using a smart lock system is a practical way to enhance overall safety procedures and strengthen office security by minimizing the risks of potential unauthorized entry.

Face recognition software and radio frequency identification (RFID) technology are two cutting-edge authentication mechanisms at the core of this proposed solution. Authorized individuals can effortlessly manage access by using RFID-enabled credentials, which enable door unlocking with just a swipe or tap (Yildirim, 2020). In addition, by confirming people's identities using distinctive facial features, facial recognition technology provides an extra degree of security (Li, 2021). By combining different authentication methods in a harmonious way, the smart lock system ensures that only authorized individuals are able to access ITDS offices, reducing the possibility of future unauthorized entrance and strengthening security measures overall (Caballero-Gil et al., 2023).

The main goal of this study is to conduct a thorough analysis of the technological needs, implementation difficulties, and possible advantages related to the installation of a smart lock system within the ITDS faculty. The proponents hoped to offer useful insights and suggestions for improving the workplace's security procedures and creating a safer, more secure atmosphere that supports creative efforts and academic achievement by

thoroughly analyzing the effectiveness of this novel security solution.

Objectives of the Study

The general objective of this study is the development of the Smart Door Lock Face Recognition and RFID Technology for ITDS Department-SC.

The specific objectives of the study include:

1. To implement a smart door lock system that will prevent potential unauthorized entry or access to in the ITDS faculty room.
2. To enhance the current security by;
 - a) Using face recognition and RFID technology which is easy to use & more secure.
 - b) Optimizing access for ITDS faculty, making it more efficient & reliable.
 - c) Ensuring portability for easy installation and maintainability for long-term usability.
3. To evaluate the performance and reliability of the proposed smart door lock system under various conditions, using the ISO/IEC 25010 standard as the basis for system evaluation.

2. RELATED LITERATURE

In today's modern academic environment, strong security protocols are essential given the speed at which technology is developing (Whitman & Mattord, 2021). Maintaining the privacy of faculty offices is essential for the preservation of precious resources as well as for creating an atmosphere that supports research, instruction, and teamwork (Garfinkel, 2017). While using traditional door locking mechanisms, the Department of Information Technology and Data Science (ITDS) at Bulacan State University Sarmiento Campus struggles with the growing threat of unauthorized access, highlighting the inherent vulnerabilities of such traditional security solutions (Sai et al., 2024).

Unauthorized entry situations involving students, staff, or other individuals not only endanger the academic spaces' integrity but also put faculty members' safety and privacy in danger. Realizing the importance that these vulnerabilities must be fixed, there is now a chance to implement a strong security solution that makes use of cutting-edge technology. Adopting a smart lock system becomes a realistic way to improve general safety procedures and strengthen office security by reducing the likelihood of unwanted entry (Caballero-Gil et al., 2023).

Modern authentication techniques, such as face recognition software and radio frequency identification (RFID) technology, are essential to this proposed solution. RFID-enabled credentials simplify door unlocking by enabling authorized users to control access using simple swipe or tap movements (Yeniman Yildirim et al., 2011). RFID systems consist of tags and readers that communicate using radio waves. The tags, embedded with a microchip and an antenna, store unique identification data. When an RFID

tag comes within the range of an RFID reader, the reader sends a signal that activates the tag. The tag then transmits its stored data back to the reader, which processes the information to grant or deny access (Weinstein, n.d.). RFID technology offers several advantages for security applications, such as quick and easy access, advanced encryption techniques to protect data, and scalability to accommodate a large number of users and access points (Nayak et al., 2015).

The approach to overcoming the flaws in traditional security measures is found in the combination of RFID technology with face recognition software. Modern authentication techniques not only improve access control effectiveness but also raise the bar for security in educational institutions. Through the use of RFID-enabled credentials, authorized personnel can move through entry points with ease, which simplifies operating procedures while upholding strict security measures (Yeniman Yildirim et al., 2011). Moreover, the integration of face recognition technology adds a further degree of verification, using the distinctiveness of each person's facial characteristics to reinforce access control protocols (Li & Jain, n.d.).

Meanwhile, facial recognition technology adds an extra level of security by confirming people's identities using distinguishing facial traits (Li, 2021). Facial recognition technology uses biometric software to map an individual's facial features mathematically and store the data as a faceprint. When a user attempts to gain access, the system captures their facial image and compares it to the stored faceprints. If a match is found, access is granted. This technology boasts high accuracy, even in varied lighting conditions and from different angles (Szeliski, 2010). It allows for seamless authentication without the need for physical contact or intervention, enhancing user convenience

and experience. Furthermore, by leveraging unique facial features that are difficult to replicate, facial recognition provides a high level of security, reducing the risk of unauthorized access (Jain et al., n.d.).

By integrating various authentication methods in a harmonious manner, the smart lock system guarantees that only authorized staff may access ITDS offices. This reduces the possibility of unauthorized entrance and strengthens security measures overall (Caballero-Gil et al., 2023). The combination of RFID technology with face recognition software addresses the flaws in traditional security measures. Modern authentication techniques not only improve access control effectiveness but also raise the bar for security in educational institutions. Through the use of RFID-enabled credentials, authorized personnel can move through entry points with ease, simplifying operating procedures while upholding strict security measures (Yeniman Yildirim et al., 2011). Moreover, the integration of face recognition technology adds a further degree of verification, using the distinctiveness of each person's facial characteristics to reinforce access control protocols (Li & Jain, n.d.).

The mutually beneficial link between these cutting-edge authentication techniques highlights the comprehensive approach used in the installation of smart lock systems. The smart lock system minimizes the danger of illegal entry and protects the integrity of academic spaces by integrating varied technologies effortlessly and allowing access to only authenticated staff (Caballero-Gil et al., 2023). Combining these technologies puts academic institutions at the forefront of innovation and resistance against ever-evolving threats while also addressing their immediate security problems and laying the foundation for future

breakthroughs in security infrastructure.

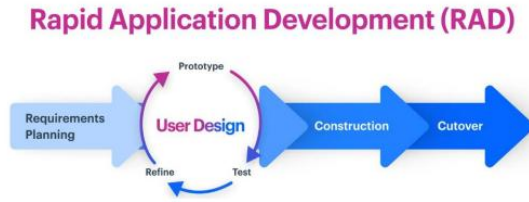
Moreover, the integration of IoT (Internet of Things) technology in smart locking systems further enhances their functionality and convenience. IoT enables seamless communication between devices, allowing for real-time monitoring and control. The use of microcontrollers, such as NodeMCU, as the central controller in these systems ensures efficient management of sensor inputs and user commands. This setup allows for features like remote access, real-time notifications, and automated locking/unlocking based on predefined schedules or triggers (Weinstein, n.d.). Additionally, cloud services facilitate data storage and synchronization across devices, providing a platform for managing user credentials, device status, and security alerts.

The implementation of smart lock systems signifies a paradigm shift in academic institutions' security procedures, moving away from traditional methods and toward a more proactive and technologically driven framework. Institutions may improve their security posture and foster an atmosphere that encourages cooperation, creativity, and academic achievement by utilizing facial recognition and RFID technologies. As a result, the implementation of smart lock systems has the potential to completely alter the security environment at educational institutions by establishing new benchmarks for effectiveness, efficiency, and flexibility in the face of changing security threats.

3. METHODOLOGY

Rapid Application Development (RAD) is a methodology that emphasizes fast and adaptable development. It involves breaking down the development process into smaller, manageable parts and gathering user feedback continuously. This method allows quick changes and ensures progress during

the course of the study. The researchers adapted this method since it is suitable and applicable for this study.



The following phases were used in determining and analyzing the process of constructing the system:

1. **Requirements Planning:** During the first phase of the study, the proponents carefully examined the issues faced by the ITDS department regarding unauthorized access to the faculty room. Information was gathered through online research, observation, and questionnaires to understand the problem better. Additionally, the researchers interviewed faculty members and administrators to learn about their needs and ideas for improving security.
2. **User Design:** This phase focused on constructing the features of the system. The primary goal is to design a user-friendly and modern structure that enhances the overall user experience. The starting point was creating an initial prototype of the system to visualize the design concepts. This prototype served as a foundational representation, allowing assessment of the system's structure and functionality.
3. **Construction:** In the third phase of the study, the system development was performed. The system design was refined, and requirements were validated. Necessary components and features for the potential operational state of the system were also finalized.
4. **Cutover:** In the final phase, the

system starts to perform some of its intended functions, marking a significant step toward operational status. This phase also includes ongoing development and continuous improvements, to ensure that the system evolves and adapts to meet the needs and requirements.

4. RESULT AND DISCUSSIONS

Using the Rapid Application Development (RAD) methodology to build the Smart Door Lock System for the ITDS faculty room, it allows us to design, test, and improve the system continuously based on feedback. We split the project into four key phases: Requirements Planning, User Design, Construction, and Cutover. Each phase helped us gather insights, build and test a prototype, and refine the system to fit the department's needs. Implementing the RAD phases helps us to quickly respond to feedback, making sure our system is both secure and user-friendly.

Data Analysis

Purposive sampling was the method utilized in this investigation to select study participants. To improve the data representation, this sampling strategy took into account every respondent's response and did not eliminate any of them.

The researcher used the Likert Scale and the following formula to interpret the survey questionnaire results:

$$AM = \sum_{i=1}^n \frac{a_i}{n} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

Where:

AM – Arithmetic Mean

a_i – The value of the *i*th observation

n – The number of observations

Table 4.1. Likert Scale Conversion Table

DESCRIPTION	VALUE	CONVERSION
Excellent (E)	5	4.51-5.00
Very Good (VG)	4	3.51-4.50
Good (G)	3	2.51-3.50
Fair (F)	2	1.51-2.50
Poor (P)	1	0.51-1.50

The system's evaluation involved 28 total respondents composed of 4 IT Professionals, 4 ITDS Professors and 20 Information Technology Students. The researchers used an ISO 25010 based Likert Scale type of evaluation tool. The evaluation assesses the system acceptability in terms of functionality suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

This study used an ISO/IEC 25010 based Likert scale type of evaluation tool. The evaluation assesses the system acceptability in terms of functionality suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

4.2. Overall Summary

Criterion	Evaluator	Mean
Functional Suitability	Professors	4.67
	Experts	4.83
	Students	4.72
Performance Efficiency	Professors	4.67
	Experts	4.83
	Students	4.98
Compatibility	Professors	4.75
	Experts	4.82
	Students	4.3
Usability	Professors	4.54
	Experts	4.28
	Students	4.72
Reliability	Professors	4.44
	Experts	4.82
	Students	4.77
Security	Professors	4.3
	Experts	4.35
	Students	4.75
Maintainability	Professors	4.7
	Experts	4.65
	Students	4.74
Portability	Professors	4.67
	Experts	4.67
	Students	4.69
Overall Mean	Experts	4.61
	Students	4.71
	Professors	4.55

System Overview

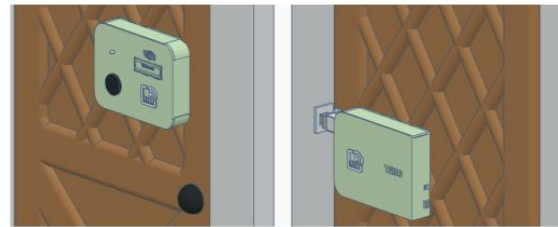


Figure 4.1: Concept Design for ITDS Smart Door Lock Arduino Device (For Implementation)

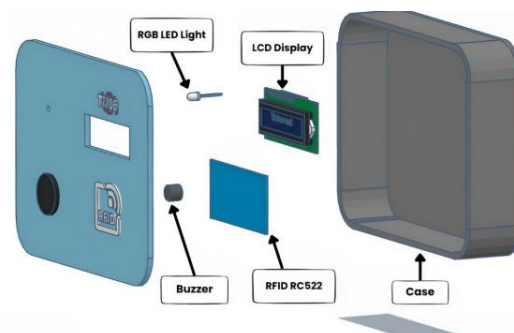


Figure 4.2: Exploded Diagram for ITDS

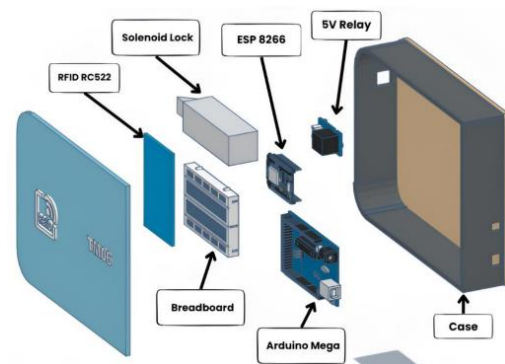


Figure 4.3: Exploded Diagram for ITDS Smart Door Lock (Back View)



Figure 4.4: Prototype (Front View)



Figure 4.5: Prototype (Back View)

5. CONCLUSION

The "Smart Door Lock Face Recognition and RFID Technology for ITDS Department-SC" successfully demonstrates an advanced approach to security through the integration of face recognition and RFID authentication. With the primary goals of preventing unauthorized access and modernizing security practices, the system effectively meets the criteria for both functional and nonfunctional requirements according to ISO/IEC 25010 standards. Areas identified

for minor improvements, such as compatibility and security robustness, provide valuable insights for future refinement. Overall, this system represents a significant step toward efficient and secure access management for institutional settings.

6. RECOMMENDATIONS

Although the Smart Door Lock system is quite effective and user-friendly, there are some areas that need improvement to maximize security and customer satisfaction. Simplifying the design of the user interface, especially by cutting down on animations on the login page, is one important suggestion. This change would improve the system's functionality and efficiency by enabling quicker access and less data usage, particularly in environments with poor connectivity. Although the current user-friendly interface is valued, these improvements could improve usability and accessibility.

Regarding security, a few enhancements are recommended. Because there is now no accountability, the use of a single generic admin account presents a serious danger. By implementing distinct admin accounts with robust password rules, it would be possible to track individual actions and lower the possibility of unwanted access. It's also critical to install multi-factor authentication (MFA) on the admin panel. In accordance with industry's best practices, this additional security measure would guarantee that unauthorized users cannot access the system, even in the event that credentials are compromised. Turning off browser password storage is advised to further secure administrator credentials. This step would strengthen the system's overall security in multi-user settings by preventing unintentional access on shared devices.

The addition of a "view picture" option for uploaded facial scans is a suggested

feature for convenience. Upon establishing the required server infrastructure, this functionality would enable administrators to confirm and authenticate facial information. Authorized personnel would benefit from an additional layer of verification and improved identity recognition accuracy with this feature. Collectively, these suggestions seek to enhance the system's use and functionality while strengthening its security.

REFERENCES

Anne Dequiros, R. C., Liezl Catacutan Jonel V Dela Cruz, C. A., Dawn Ziganay, K. G., & Guarin, B. M. (n.d.). *TRENDSAFE: Intrusion Alarm System for Smart Home Protection and Security Using Arduino with SMS Notification*.

Chakraborty, A., Islam, M., Shahriyar, F., Islam, S., Zaman, H. U., & Hasan, M. (2023). Smart Home System: A Comprehensive Review. In *Journal of Electrical and Computer Engineering* (Vol. 2023). Hindawi Limited.
<https://doi.org/10.1155/2023/7616683>

Domdouzis, K., Kumar, B., & Anumba, C. (2007). Radio-Frequency Identification (RFID) applications: A brief introduction. *Advanced Engineering Informatics*, 21(4), 350–355.
<https://doi.org/10.1016/j.aei.2006.09.001>

Garfinkel, Simson., Spafford, Gene., & Schwartz, A. (n.d.). *Practical UNIX and Internet security*.

Li, S. Z., & Jain, A. K. (n.d.). *Handbook of Face Recognition (The Second Edition)* Springer-Verlag.

Okowa, E., Elechi, P., & Ekwueme, U. (n.d.). Facial Recognition Based Smart Door Lock System. *FUPRE Journal of Scientific and Industrial Research*, 6(2), 2022.
<https://www.researchgate.net/publicatio>

n/360783137

Paranagama, C., Hettige, B., & Paranagama, P. (n.d.). *A Review on Existing Smart Door Lock Systems*.
<https://doi.org/10.13140/RG.2.2.18892.08325>

Perfecto R. Ruaya, Jr. (2023). Smart Lock Technology: Developing and Enhancing Home Security using Android-Based Controlled Door Locking App's. *International Journal of Advanced Research in Science, Communication and Technology*, 538–547. <https://doi.org/10.48175/ijarsct-12176>

Whitman, M. E. (2011). *Principles of Information Security Fourth Edition*.

Elechi, P., Ekwueme, U., & Okowa, E. (2022, May). *Facial Recognition Based Smart Door Lock System* [Review of *Facial Recognition Based Smart Door Lock System*]. *Journal of Scientific & Industrial Research* 6(2):95–105.

Zedig, J. (2022). *Security in Smart Locks: Do Consumers Care?* [Review of *Security in Smart Locks: Do Consumers Care?*]. Digitala Vetenskapliga Arkivet; Independent thesis Basic level (university diploma).

Ruaya, P. R. (2023, July). *Smart Lock Technology: Developing and Enhancing Home Security using Android-Based Controlled Door Locking App's* [Review of *Smart Lock Technology: Developing and Enhancing Home Security using Android-Based Controlled Door Locking App's*]. *International Journal of Advanced Research in Science Communication and Technology*. 10.48175/IJARSCT-12176.

Rosli, A. A., Razif, M. R. M., Hassan, O. A., Shah, N. S. M., Nordin, I. N. A. M., & Mustafa, K. N. (2021). Development of Smart Home System based on Mobile Apps Control Using IoT for Educational Purposes. *Alinteri Journal of Agriculture Sciences*, 36(1), 525–

533.

<https://doi.org/10.47059/alinteri/v36i1/ajas21075>

Saumat, M. T., Herc, C., Codate, C., & Dayoc, R. P. (n.d.). *User-Friendly Smart Security System using ESP32 in Rental Rooms*. www.sajst.org

Aisyah, S., Ali, Y., Saharja, K., Suhendra, S., & Sani, A. (2022). SMART DOOR LOCK SYSTEM DEVELOPMENT PROTOTYPE USING RFID TECHNOLOGY ID-12. *Jurnal Riset Informatika*, 4(4), 379–384. <https://doi.org/10.34288/jri.v4i4.433>

Almario, E. T., Dominique, J., Beltran, T., Benne, N., Besin, M., Alvin, E., & Mancilla, M. (n.d.). *COMPUTER ENGINEERING CRIME TRACKING DEVICE USING GPS AND SMS TECHNOLOGIES* Researchers: Clarizza Muriel.

Caballero-Gil, C., Álvarez, R., Hernández-Goya, C., & Molina-Gil, J. (2023). Research on smart-locks cybersecurity and vulnerabilities. *Wireless Networks*. <https://doi.org/10.1007/s11276-023-03376-8>

Dela Cerna, R., Montesclaros, A., James Ruiz, S., Amalie Quiñones, C., Villanueva, C., Van Villena, I., & Paolo Gumia, N. (n.d.). *UNIFIED ATTENDANCE MoNIToRiNG UsiNG rFID*.

Francis, J., Barcibal, A., Orejudos, D. J., Christopher, C., Taer, B., & Capricho, N. C. (n.d.). *THE OFFICIAL JOURNAL OF ENGINEERING RESEARCH*.

Jain, A. K., Ross, A. A., & Nandakumar, K. (n.d.). *Introduction to Biometrics*.

Luna, A. A., Christian Reyes Adviser, R. R., & Liza Marie Nuevo, E. (n.d.). *INNOVATIVE ENTRANCE GATE WITH METAL DETECTOR APPLYING FACE RECOGNITION DETECTION AND NOTIFICATION SYSTEM* Researchers: Ramwell T.

Nayak, R., Singh, A., Padhye, R., & Wang, L. (2015). RFID in textile and clothing manufacturing: technology and challenges. In *Fashion and Textiles* (Vol. 2, Issue 1). Springer Singapore. <https://doi.org/10.1186/s40691-015-0034-9>

Sai, K. N., Sunil, Dr. T., & Eshwarappa, Dr. M. (2024). A comprehensive review of door lock security systems. *International Journal of Circuit, Computing and Networking*, 5(1), 12–17. <https://doi.org/10.33545/27075923.2024.v5.i1a.61>

Szeliski, R. (2010). *Computer Vision: Algorithms and Applications*. <http://szeliski.org/Book/>.

Verma, G. K., & Tripathi, P. (2010). A Digital Security System with Door Lock System Using RFID Technology. *International Journal of Computer Applications*, 5(11), 6–8. <https://doi.org/10.5120/957-1334>

Weinstein, R. (n.d.). *RFID: A Technical Overview and Its Application to the Enterprise*. <http://www.rfiduk>.

Yeniman Yildirim, E., Akalp, G., Aytac, S., & Bayram, N. (2011). Factors influencing information security management in small- and medium-sized enterprises: A case study from Turkey. *International Journal of Information Management*, 31(4), 360–365. <https://doi.org/10.1016/j.ijinfomgt.2010.10.006>

